

RISHITHA CHILAKA

AI/ML Engineer | RAG & LLM Applications | Applied Machine Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience building and productionizing machine learning, Generative AI, and data-driven applications. Experienced in building enterprise Retrieval-Augmented Generation (RAG) systems, LLM applications, retrieval pipelines, vector search, document processing, model evaluation, and hallucination mitigation. Strong hands-on expertise with Python, PyTorch, Azure OpenAI, Hugging Face Transformers, pgvector, MLflow, Docker, GitHub Actions, and Microsoft Azure.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Bash, Git, Pandas, NumPy **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Large Language Models(LLMs), Re-Ranking, Vector embeddings, Semantic search, Prompt engineering, Azure OpenAI, Hugging Face Transformers, Hallucination mitigation **LLM Evaluation:** Golden datasets, Evaluation harnesses, Retrieval recall@k, Groundedness/Faithfulness evaluation, Regression testing, Error analysis, Prompt versioning **Machine Learning:** PyTorch, Scikit-learn, NLP, Document Classification, Feature Engineering, Classification, model evaluation **AI/ML systems & MLOPs:** Vector Databases, pgvector, Document Processing, OCR, Retrieval Pipelines, MLflow, CI/CD, GitHub Actions, Docker, Model Monitoring, Model Versioning, Rollback **Cloud:** Microsoft Azure

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Optum — Houston, TX | Jan 2025 - Present

- Architected and productionized components of an enterprise Retrieval-Augmented Generation (RAG) platform using Azure OpenAI, Hugging Face Transformers, vector embeddings, and pgvector to support retrieval and synthesis across 40,000+ enterprise documents and structured/unstructured data.
- Engineered a distributed Python ETL and document-processing pipeline for parsing, OCR fallback, section-aware chunking, embedding generation, and incremental vector re-indexing, supporting continuous ingestion and updates across a large enterprise document corpus.
- Redesigning document chunking around section headings and introducing a re-ranking stage, increasing retrieval Recall@10 from 0.58 to 0.79 and improving the quality of downstream LLM responses.
- Integrated golden-dataset evaluation into GitHub Actions CI/CD to measure groundedness and Recall@K while catching 3 regressions before production.
- Implemented out-of-corpus question detection and analysts routing rather to prevent unsupported responses, reducing evaluated hallucination rate and improving answer reliability on the evaluation dataset.
- Improved document-classification F1 from 0.61 to 0.83 across 14 document classes by developing and productionizing a PyTorch-based classifier, replacing keyword-based routing for enrollment paperwork.
- Implemented prompt versioning, change controls, and rollback mechanisms after identifying a two-day answer-quality regression caused by an untracked prompt change, strengthening production reliability and recovery capabilities.

Machine Learning Engineer

IBM — Bangalore | Jan 2020 - Jul 2022

- Developed anomaly-detection models for log-processing workloads handling 450K+ logs per hour, contributing to a 32% reduction in false positives.
- Developed and evaluated NLP classification models using spaCy, improving classification accuracy by 26% across supported workloads.
- Optimized TensorFlow- and CUDA-based inference pipelines to improve model response time and support efficient ML model serving.
- Implemented data-drift monitoring processes to identify changes in production data distributions and support ongoing model stability and performance.
- Supported deployment and maintenance of containerized machine learning services using Python and ML technologies.
- Applied feature engineering, model evaluation, and error analysis to identify performance bottlenecks and improve the reliability of machine learning models.

EDUCATION

Union Commonwealth University — Master of Science, Management Information Systems

Aug 2022 - Dec 2024